

Solution Sheet-06

1. We use the identity $\max(a, b) = \frac{a+b+|a-b|}{2}$.

$$E[\max(X^2, Y^2)] = E\left[\frac{X^2 + Y^2 + |X^2 - Y^2|}{2}\right] = \frac{1}{2}(E[X^2] + E[Y^2] + E[|X^2 - Y^2|]).$$

Given X, Y have means 0 and variances 1, we have $E[X^2] = \text{Var}(X) + (E[X])^2 = 1$. Similarly, $E[Y^2] = 1$.

$$E[\max(X^2, Y^2)] = \frac{1}{2}(1 + 1 + E[|X^2 - Y^2|]) = 1 + \frac{1}{2}E[|X^2 - Y^2|].$$

Now consider the term $|X^2 - Y^2| = |(X - Y)(X + Y)|$. By the Cauchy-Schwarz inequality for expectations ($E[|UV|] \leq \sqrt{E[U^2]E[V^2]}$):

$$E[|(X - Y)(X + Y)|] \leq \sqrt{E[(X - Y)^2]E[(X + Y)^2]}.$$

Calculate the second moments:

$$\begin{aligned} E[(X - Y)^2] &= E[X^2 - 2XY + Y^2] = 1 - 2\rho + 1 = 2(1 - \rho). \\ E[(X + Y)^2] &= E[X^2 + 2XY + Y^2] = 1 + 2\rho + 1 = 2(1 + \rho). \end{aligned}$$

Substituting these back:

$$E[|X^2 - Y^2|] \leq \sqrt{2(1 - \rho) \cdot 2(1 + \rho)} = \sqrt{4(1 - \rho^2)} = 2\sqrt{1 - \rho^2}.$$

Finally:

$$E[\max(X^2, Y^2)] \leq 1 + \frac{1}{2}(2\sqrt{1 - \rho^2}) = 1 + \sqrt{1 - \rho^2}.$$

2. The PDF is uniform ($f(x, y) = 1/2$) over the region $D = \{(x, y) : |x| + |y| \leq 1\}$.

(i) Independence: For independence, the joint PDF must factorize as $f(x, y) = f_X(x)f_Y(y)$, which implies the support region must be a rectangle (product of intervals). Here, the range of x depends on y (specifically, $|x| \leq 1 - |y|$). Since the domain is not rectangular, X and Y are not independent.

(ii) Correlation: Check the covariance $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$. The region D is symmetric with respect to both the x-axis and y-axis.

$$E[X] = \iint_D xf(x, y) dx dy = \frac{1}{2} \int_{-1}^1 \int_{-(1-|x|)}^{1-|x|} x dy dx = 0 \quad (\text{odd function in } x).$$

Similarly, $E[Y] = 0$.

$$E[XY] = \frac{1}{2} \iint_D xy dy dx.$$

Since the integrand xy is an odd function with respect to x (and y) and the domain is symmetric, the integral is 0. Thus, $\text{Cov}(X, Y) = 0$, implying X and Y are uncorrelated.

3. Given X_1, X_2 are independent symmetric uniform variables. Mean $E[X_1] = 0, E[X_2] = 0$. Variance of $U\{-k, \dots, k\}$ is $\sigma_k^2 = \frac{k(k+1)}{3}$. So, $\sigma_1^2 = \frac{n(n+1)}{3}$ and $\sigma_2^2 = \frac{m(m+1)}{3}$.

Define $Z = cX_1 + sX_2$ and $W = cX_2 - sX_1$, where $c = \cos \theta, s = \sin \theta$.

Covariance:

$$\begin{aligned} \text{Cov}(Z, W) &= E[(cX_1 + sX_2)(cX_2 - sX_1)] \\ &= c^2 E[X_1 X_2] - cs E[X_1^2] + sc E[X_2^2] - s^2 E[X_1 X_2] \\ &= 0 - cs\sigma_1^2 + sc\sigma_2^2 - 0 \quad (\text{by independence}) \\ &= sc(\sigma_2^2 - \sigma_1^2) = \frac{1}{2} \sin(2\theta)(\sigma_2^2 - \sigma_1^2). \end{aligned}$$

Variances: $\text{Var}(Z) = c^2\sigma_1^2 + s^2\sigma_2^2$ and $\text{Var}(W) = c^2\sigma_2^2 + s^2\sigma_1^2$.

Correlation Coefficient:

$$\rho_{Z,W} = \frac{\frac{1}{2} \sin(2\theta)(\sigma_2^2 - \sigma_1^2)}{\sqrt{(c^2\sigma_1^2 + s^2\sigma_2^2)(c^2\sigma_2^2 + s^2\sigma_1^2)}}.$$

The magnitude $|\rho|$ is maximized when $\theta = \pi/4$, where $\text{Var}(Z) = \text{Var}(W) = \frac{\sigma_1^2 + \sigma_2^2}{2}$.

$$|\rho_{Z,W}| \leq \frac{\frac{1}{2}|\sigma_2^2 - \sigma_1^2|}{\frac{\sigma_1^2 + \sigma_2^2}{2}} = \frac{|\sigma_2^2 - \sigma_1^2|}{\sigma_1^2 + \sigma_2^2}.$$

Substituting $\sigma_1^2 = \frac{n(n+1)}{3}$ and $\sigma_2^2 = \frac{m(m+1)}{3}$, the factor $1/3$ cancels:

$$|\rho_{Z,W}| \leq \frac{|m(m+1) - n(n+1)|}{m(m+1) + n(n+1)}.$$

4. (a) There are n cells. The probability of a ball falling into a specific cell is $p = 1/n$. Since there are r balls distributed independently, this is a sequence of r Bernoulli trials. Thus, $X \sim \text{Binomial}(r, 1/n)$.

$$p_k = P(X = k) = \binom{r}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{r-k}.$$

(b) We are given $r, n \rightarrow \infty$ such that $r/n = \lambda$ (constant). Thus $1/n = \lambda/r$. Substitute this into the PMF:

$$\begin{aligned} p_k &= \frac{r(r-1)\dots(r-k+1)}{k!} \left(\frac{\lambda}{r}\right)^k \left(1 - \frac{\lambda}{r}\right)^{r-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{r}{r} \frac{r-1}{r} \dots \frac{r-k+1}{r}}_{\rightarrow 1} \cdot \underbrace{\left(1 - \frac{\lambda}{r}\right)^r}_{\rightarrow e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{r}\right)^{-k}}_{\rightarrow 1}. \end{aligned}$$

Taking the limit as $r \rightarrow \infty$:

$$\lim_{r \rightarrow \infty} p_k = \frac{\lambda^k e^{-\lambda}}{k!}.$$

5. Consider the ratio of consecutive terms in the Poisson PMF:

$$\frac{P(X = k)}{P(X = k - 1)} = \frac{e^{-\lambda} \lambda^k / k!}{e^{-\lambda} \lambda^{k-1} / (k-1)!} = \frac{\lambda}{k}.$$

The probability increases ($P(k) > P(k-1)$) if $\frac{\lambda}{k} > 1 \implies k < \lambda$. The probability decreases ($P(k) < P(k-1)$) if $\frac{\lambda}{k} < 1 \implies k > \lambda$.

Case 1: λ is not an integer. The sequence increases for all integers $k < \lambda$ and decreases for all $k > \lambda$. The peak is at the largest integer $k < \lambda$, i.e., $k = \lfloor \lambda \rfloor$.

Case 2: λ is an integer. At $k = \lambda$, the ratio is exactly 1, meaning $P(\lambda) = P(\lambda - 1)$. Thus, there are two maximum values (modes) at $k = \lambda - 1$ and $k = \lambda$.

6. (a) We want $P(X_1 = k_1, \dots, X_n = k_n \mid S_n = m)$. Note that the event $\{X_1 = k_1, \dots, X_n = k_n\}$ implies $S_n = \sum k_i = m$.

$$P(\mathbf{X} = \mathbf{k} \mid S_n = m) = \frac{P(X_1 = k_1, \dots, X_n = k_n)}{P(S_n = m)}.$$

Since X_i are independent, the numerator is $\prod_{i=1}^n \frac{e^{-\lambda} \lambda^{k_i}}{k_i!} = e^{-\Lambda} \frac{\prod \lambda^{k_i}}{\prod k_i!}$. Since $S_n \sim P(\Lambda)$, the denominator is $\frac{e^{-\Lambda} \Lambda^m}{m!}$.

$$\text{Ratio} = \frac{e^{-\Lambda} \prod \lambda^{k_i} / \prod k_i!}{e^{-\Lambda} \Lambda^m / m!} = \frac{m!}{k_1! \dots k_n!} \frac{\lambda_1^{k_1} \dots \lambda_n^{k_n}}{\Lambda^m}.$$

Rewrite $\Lambda^m = \Lambda^{\sum k_i} = \prod \Lambda^{k_i}$:

$$= \frac{m!}{k_1! \dots k_n!} \prod_{i=1}^n \left(\frac{\lambda_i}{\Lambda} \right)^{k_i}.$$

7. (a) Let X be vehicles from Road A ($\lambda_A = 2$) and Y from Road B ($\lambda_B = 3$). Given $S = X + Y = 10$. From standard results (or Q6 with $n = 2$), $X \mid S = 10 \sim \text{Binomial}(n = 10, p)$.

$$p = \frac{\lambda_A}{\lambda_A + \lambda_B} = \frac{2}{2 + 3} = 0.4.$$

So, $X \mid S = 10 \sim \text{Binomial}(10, 0.4)$.

$$(b) P(X = 4) = \binom{10}{4} (0.4)^4 (0.6)^6.$$

$$= 210 \times 0.0256 \times 0.046656 \approx 0.2508.$$

- (c) Let Revenue $R = 50X + 80Y$. Since $X + Y = 10$, we have $Y = 10 - X$.

$$R = 50X + 80(10 - X) = 800 - 30X.$$

We need the expected revenue given $n = 10$:

$$E[R \mid n = 10] = 800 - 30E[X \mid n = 10].$$

The mean of a Binomial(10, 0.4) is $np = 10(0.4) = 4$.

$$E[R \mid n = 10] = 800 - 30(4) = 800 - 120 = \text{Rs. } 680.$$

8. (a) Rate $\lambda = 3$ calls/min. For $t = 2$ mins, parameter $\mu = \lambda t = 6$.

$$P(N(2) = 5) = \frac{e^{-6}6^5}{5!} \approx 0.1606.$$

- (b) For $t = 30$ sec (0.5 min), parameter $\mu = 3 \times 0.5 = 1.5$. We want the probability of zero calls:

$$P(N(0.5) = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} \approx 0.223.$$

- (c) Case 1 (20 sec): $t = 1/3$ min. $\mu = 3(1/3) = 1$.

$$P(1 \text{ call}) = \frac{e^{-1}1^1}{1!} = e^{-1} \approx 0.3679.$$

- Case 2 (40 sec): $t = 2/3$ min. $\mu = 3(2/3) = 2$.

$$P(2 \text{ calls}) = \frac{e^{-2}2^2}{2!} = 2e^{-2} \approx 0.2707.$$

Conclusion: It is more likely to receive exactly 1 call in 20 seconds ($0.368 > 0.271$).